

Envisioning excellence in the Affect of Service, Information Control and Library as a Place: a LibQUAL+® assessment of Pakistani School Libraries

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Abstract

Purpose – The purpose of this study is to examine the perceived, minimum acceptable, and desired service quality of School Libraries in terms of the Affect of Service, Information Control, and as a Place using a unique LibQUAL+® Urdu-translated version.

Design/methodology/approach – A quantitative survey research method and an instrument tool were used to conduct this study. The already available Urdu-translated version (translated for university library users) was content-validated and reliability-checked on 9th and 10th-grade students through two phases of pilot testing, content validation, and incorporation of suggestions. Then, for the final study, data were collected from 28 public and private sector school students in seven Tehsils of Sargodha District using equal-sized stratified convenience sampling.

Findings – Descriptive statistics show that the perceived service quality in Private and Public Schools exceeds the middle value of 4.5 on the 9-point Likert-type scale. However, students' minimum acceptable and desired service quality are higher than the perceived quality in the Affect of Service, and Information Control categories. Library as a Place, however, scored higher than the minimum acceptable service quality. Inferential statistics reveal that male respondents rate the library quality significantly higher than female respondents. The results also indicate that the minimum acceptable level of library quality reported by government school respondents is significantly higher than that of private school students. Male respondents have reported significantly higher expectations than female respondents.

Originality/value – This study is the first to use the LibQUAL+® tool's Urdu language version to investigate the libraries' School Libraries service quality in Pakistan.

Keywords LibQUAL+®, Service quality, School Libraries, Affect of Service, Information Control, Library as a Place

Paper type Research paper

Introduction

Academic libraries play an essential role in educational institutions by supporting students' learning and academic development (Dahan *et al.*, 2016). These academic libraries are divided into three types – university, college, and school libraries. School libraries serve as students' first exposure to a library environment. According to the American Library Association (ALA), a school library provides access to a wide range of resources and a setting where teaching and learning are prioritized. It offers a tailored space and setting for students' academic achievement, helping them enhance literacy and learning outcomes (Lance and Hofschire, 2011).

A review of the existing literature indicated that university libraries have been extensively researched using different service quality (SQ) measures (primarily SERVQUAL and LibQUAL+®) internationally (Alam, 2021; Kekana and Kheswa, 2022; Wang *et al.*, 2022) and within Pakistan (Ahmed, 2017; Batool *et al.*, 2020).



The literature reflects a similar situation regarding college libraries' SQ (Ata *et al.*, 2019; Ghaedi *et al.*, 2020; Mallya and Patwardhan, 2018). However, school library quality research is still in its infancy worldwide. The literature review identifies three major research gaps at both the international and national levels. First, there is limited literature on school library SQ worldwide. Second, no study has validated or reliability-checked the Urdu-translated LibQUAL+® tool for the School Libraries on a sufficiently representative sample. Third, no research has examined school library SQ using any standardized tool in Pakistan, leaving a significant research gap.

Therefore, this research aimed to bridge the literature gap using the Urdu-translated LibQUAL+® tool, which had never been applied to Pakistani School Libraries before. This study is also unique in its sample, focusing on public and private school students.

Statement of the problem

Kim (2009) noted that school libraries have been under-researched, and emphasized the need to fill this gap. Present literature search conducted 16 years after Kim's claim confirmed this gap still exists. Second, Creswell (2013) stated that overlooked methodologies in previous studies also constitute research gaps. The literature review reveals that no international or national study has examined school library SQ using the LibQUAL+® tool's Urdu translated version. This highlights a methodological gap.

Third, school library SQ has never been systematically investigated in Pakistan. No empirical data exists to guide librarians, school administrators, or policymakers about the current status of school libraries regarding services, information access, and infrastructure. Therefore, this study is designed to fill the research gap, validate and reliability-check the Urdu language version of the LibQUAL+® tool in school libraries, and provide suggestions to the school governing bodies, school librarians, and administration for improving their library services.

Objectives of the study

The objective of the study was to answer the following research questions:

- RQ1. What is the status of perceived, minimum acceptable, and desired SQ in the realm of Affect of Service (AS); Information Control (IC); and Library as a Place (LP) at School Libraries of Sargodha?
- RQ2. What is the difference between perceived, minimum acceptable, and desired SQ in AS; IC; and LP at the School of Sargodha?
- RQ3. What is the difference in perceptions, minimum and maximum expectations in the realm of AS; IC; and LP based on the type of school (government, private), gender (male, female), and socio-cultural status (urban, rural)?

Delimitations of the study

The study examines perceived, minimum acceptable, and maximum desired library quality regarding AS, ICASIC, and LP in LP private and public-School Libraries for male and female students across seven Tehsils of Sargodha District. The investigation employs the translated Urdu version of the LibQUAL+® tool. Having established the introduction to the topic, research objectives, and limitations of the study, the following section reviews existing literature on SQ and related assessment tools.

Literature review

Since the literature on the investigation of school libraries' SQ using the LibQUAL+® tool was unavailable, general literature related to the SQ assessment of college and university

libraries has been reviewed to understand the phenomenon better so that the present investigation could be well-structured.

Service quality

A literature review for the present investigation indicated several SQ assessment research studies. Libraries function as knowledge powerhouses, providing multiple commodities and services to their academic community, including students, teachers, and administrators. Moreover, their primary goal is to meet their patrons' evolving needs and expectations while engaging in academic research activities (Mohindra and Kumar, 2015). In a library setting, quality refers to providing services efficiently and without flaws (Thapisa and Gamini, 1999). It is mainly tied to the physical infrastructure of buildings and equipment, the materials offered, staff responsiveness to inquiries, and service performance patterns. Several standardized tools have been used to gauge library SQ, such as LibQUAL+®, SERVQUAL, SERVPERF, Importance-Performance Analysis, surveys, library standards, certification, and other approaches.

Service quality of school libraries

Saufa (2023) assessed the school libraries in Yogyakarta, Indonesia, by collecting data from 93 school-grade respondents using the same LibQUAL+® tool. Three main dimensions of the SQ (AS; IC; and LP) were accessed on the perceived, minimum acceptable, and desired library SQ. The study lacked an address to interpretation bias and had not content validated or reliability checked the instrument on students of school grades. The results revealed that students were satisfied with interaction with the librarians but showed concerns related to IC. An instance of using the SERVQUAL tool to access the library SQ was found at the Largo High School, Philippines. The SERVQUAL tool, however, gauges the library SQ based only on perceptions and expectations. The study identified accessibility issues and suggested to improve adaptive services.

Along with these results based on data from the general population of school students, a set of hearing-impaired students was also investigated in Metro Manila by Gelvezon (2007). The researcher used interview and policy analysis methods to collect the data and suggested the importance of special library services for special school students. These studies reinforce the importance of library assessments using LibQUAL+® and SERVQUAL tools in investigating the service quality of school libraries.

LibQUAL+®

The LibQUAL+® instrument is used in the present investigation for its suitability and comprehensiveness, and because it investigates service quality across perceived, minimum acceptable, and desired service quality. The Association of Research Libraries (ARL) originally developed the instrument in October 1999. It has been tested in libraries worldwide since its inception, and it may be used online or in print format (Heath and Cook, 2001; Pedramnia et al., 2012; Ramezani et al., 2018). Two versions of the LibQUAL+® tools are available for evaluating library SQ: (1) LibQUAL+® and (2) LibQUAL Lite. The former is a comprehensive version, and the latter is a condensed version. The full version has 22 main questions divided into three key aspects: "AS, IC, and LP." These dimensions may be defined as follows:

Affect of Service: This component includes nine questions addressing library personnel's responsiveness, confidence, and empathy.

"Information Control" (IC): This dimension comprises eight questions examining the adequacy and timeliness of the library's materials and services.

"Library as a Place" (LP): This component consists of five questions designed to collect user feedback on infrastructure, physical space, and comfort.

This unique and powerful tool helps libraries reimagine their service offerings by identifying differences in perceived, minimum acceptable, and desired service quality (Asemi *et al.*, 2010; Khaola and Mabilikoane, 2015). This instrument is vastly used, and according to LibQUAL official website, data from one million library users across 1,000 institutions in more than two dozen countries have been collected using this instrument so far. This instrument has been translated into more than a dozen languages, including Urdu (<https://www.libqual.org/>).

This comprehensive version of the instrument is used in present investigation for two reasons. First, it includes elements of all previously developed tools arranged into the three core aspects. Second, it was already translated into the native language of the population of the present study, Urdu. Rehman (2012) in his doctoral research, translated the LibQUAL+® tool into Urdu and then validated it in university settings twice: first, in a pilot test with 75 undergraduate and postgraduate students and faculty members, and then in a final study with 560 respondents. Testing the Urdu version confirmed its validity. The condensed version of LibQUAL+® is LibQUAL Lite.

Review of LibQUAL+® studies

The LibQUAL+® tool is that unique and one of its kind that special journal issues have been published focusing on LibQUAL+®. A special Performance Measurement and Metrics issue, titled “The Maturation of Assessment in Academic Libraries: The Role of LibQUAL,” examined LibQUAL+® survey results and their implementation in ARL and non-ARL libraries. Another journal, “Journal of Library Administration,” also published a special issue containing articles on the application of LibQUAL+® surveys, later compiled into a book edited by Heath *et al.* (2014). Along with these journal issues, research studies have been conducted using the LibQUAL+® tool to assess SQ gaps between the current status, minimum acceptable, and desired SQ levels.

A larger quantity of literature in this regard comes from university settings. At Washburn University, Dole (2002) presented a LibQUAL+® pilot study. According to the results, users needed an improved library facility, updated technology, remote access to Internet information, and supportive employees. McNeil and Giesecke (2002) analysed surveys conducted in the U.S. based universities using the LibQUAL+® tool. Their findings revealed that UNL libraries performed well. Using the same tool, Hitchingham and Kenney (2002) found that graduate students at Virginia University were more satisfied with library SQ than undergraduates and professors.

After gaining traction in Western institutions, the LibQUAL+® tool became popular in the East. Asemi *et al.* (2010) investigated Iranian academic libraries using LibQUAL+® and found that respondents’ perceptions of the IC dimension were higher than their perceptions of AS and LP. Pourahmad *et al.* (2016) used a survey approach with LibQUAL to analyse the quality of services at four university libraries in North Khorasan, Iran. The study found that these libraries failed to meet student expectations. However, Ramezani *et al.* (2018) reported that users in Iranian university libraries were generally satisfied with SQ.

In India, Kumar and Mahajan (2019) found that the library at Allama Iqbal Library did not meet the minimum acceptable or desired service levels. According to them, few LibQUAL+® research studies have been conducted in India. Rehman (2013) found that university libraries did not meet their clients’ minimum or maximum SQ requirements in Pakistan. Further, the IC dimension was significantly lower than the other domains and fell short of the minimum acceptable and desired service levels.

In conclusion, several LibQUAL+® surveys have been conducted in various libraries worldwide. These studies used quantitative and mixed-methods approaches. Furthermore, research studies have been undertaken in Eastern and Western cultures, evaluating university libraries. The findings of these studies suggest that LibQUAL+® is a widely accepted and standardized tool for measuring library SQ (Cook *et al.*, 2002; Voorbij, 2012; Wei *et al.*, 2005).

Building on these theoretical foundations, this study adopts a quantitative approach to measure school library SQ. The details are in the following methods section.

Methods

Research design

A quantitative survey research design provided the methodological grounds for conducting this research.

Measurement scale

The LibQUAL+® measurement tool was used to conduct this study. The Association for Research Libraries created the LibQUAL+® tool in partnership with Texas Agricultural and Mechanical University in 1999. The United States Department of Education funded the project. The tool assesses library services by asking 22 basic questions across three dimensions – AS, IC, and LP. The instrument for the present study was used with permission from its originating body. This tool was available in many languages, including Urdu, which was translated into Urdu in a doctoral by [Rehman \(2012\)](#).

The Urdu language version was chosen for the present investigation, considering 9th and 10th grade students have better Urdu skills than English. The respondents were native Urdu/Punjabi speakers. However, translation and using a university level instrument on a sample of 9th and 10th grade students could trigger translation and interpretation biases, which were properly addressed.

Bias consideration. The LibQUAL+® tool was initially designed for university libraries and translated into Urdu from English. Applying it to 9th and 10th-grade students might introduce interpretation and translation biases. Interpretation bias often occurs when respondents perceive and respond to information differently due to their age, background, literacy level, culture, language, or context ([Hirsch et al., 2019](#); [Mathews and MacLeod, 2005](#)). The translation of the research instrument might also introduce cultural and measurement biases ([Behling and Law, 2000](#); [Van de Vijver and Poortinga, 1997](#)). Previous researchers described strategies to reduce these biases through content validity, reliability checking, and pilot testing ([Harkness et al., 2003](#); [Maneesriwongul and Dixon, 2004](#); [Sousa and Rojjanasrirat, 2011](#)). All these measures that mitigate interpretation and translation biases, including content validity, reliability, and pilot testing, were taken.

Content validation and pilot testing of the measurement scale. Once content was validated, the instrument was sent to the respondents for pilot data collection. During pilot testing, respondents initially struggled to understand the survey instrument. To address this issue, respondents were asked to reveal difficulties. The respondents' hazards were noted. Then, the researchers identified the two teachers who also taught the students about using the school library. The instrument and respondents' hazards were shown to them. In light of the respondents' and their library teachers' suggestions, the present researchers addressed all the difficulties that respondents had faced and then restructured the Urdu language version of the instrument. The refined version of the instrument was again pilot tested, but students again showed some concerns, which were addressed again. These final concerns of the students changed the scale pattern. The respondents and their teachers' final suggestions converted 22 statements (three Likert-type scales gauging perceived, minimum, and desired SQ in front of each statement) to 66 (separating each scale and 22 statements of each construct in sequence, totalling 66). Then, pilot testing data were collected again, and reliability was checked. The instrument was found reliable on a sample of school students.

Sampling procedures and techniques

Equal-sized stratified convenience sampling was used to conduct this study. Initially, the number of Tehsils in the Sargodha District was identified. Those were seven in number.

Then, two private and two Government Schools per Tehsil were selected out of seven. Out of those, one government and one private were selected for male and one each for female students (co-education at the school level in Pakistan is still rare). Then, 15 questionnaires were distributed in each of the four selected schools across each Tehsil of Sargodha District.

Justification of the equal-sized stratified convenience sampling. Equal-sized stratified convenience sampling was used in this study to reach meaningful inferences across different government/private school types and male/female gender groups.

Need for representation across key groups. The study sample comprised diverse characteristics (government/private, male/female). Therefore, it was necessary to gain representation from all these. Consequently, it was decided that the representation of all the groups would be assured in data collection in equal number. It was quite necessary in the Pakistani school education system, where there are separate schools for girls and boys and two educational systems exist: Government Schools and Private Schools.

Equal-sized stratification for comparative analysis. A balanced data from all the groups generated a need for an equal-sized data collection. Therefore, an equal representation was required from the sample of each group in equal proportion. Stratified sampling ensured that each sub-group was equally represented. Therefore, the total required questionnaires were divided equally among all the sub-groups where needed based on gender and type of school.

Feasibility and accessibility. Convenience sampling within the already selected sets of strata allowed the researchers to collect data from available respondents because diversity was already ensured earlier when making sets of strata. These methods reduced the risk of disproportionate representation, which could lead to an imbalance in sub-groups.

Data analysis – parametric vs non-parametric statistics

[Norman \(2010\)](#) examined the literature published over the last 80 years. The researcher concluded that “parametric statistics can be used with Likert data, small sample sizes, unequal variances, and non-normal distributions without fear of coming to the wrong conclusion.” Following his argument, the present researchers applied parametric statistics to the collected data.

Reliability statistics of the final study

The instrument’s reliability was rechecked through Cronbach’s alpha calculation after the final data collection. The instrument demonstrated a high reliability among school students. The reliability statistics are shown in the following [Table 1](#):

A Cronbach’s Alpha value exceeding 0.70 indicates strong internal consistency ([Nunnally, 1978](#)). Other researchers also agreed and considered values more than 0.70 reliable for an instrument’s internal consistency ([Awan et al., 2021, 2022](#)). Applying these research methods, the following section of the research presents the results of the study.

Table 1. Reliability statistics of the final study

No. of items	Cronbach’s alpha
66	0.884
Source(s): Cronbach’s Alpha – authors’ own work	

Data analysis and results

The data were analysed using SPSS 26 and using descriptive and inferential statistical tests. The demographic characteristics of the respondents were as follows:

Table 2 describes the data of 400 responses. Among these, 209 (52.3%) were males, and 190 (47.5%) females. One respondent preferred not to reveal the gender. The sample showed a sufficient representation of rural and urban students, making 189 (47.3%) responses from rural residents and 211 (52.8%) from urban residents. The 229 (57.3%) respondents were from Private Schools; and the rest, 165 (41.3%), were from Government Schools.

The following calculations revealed the mean and standard deviation of the perceived SQ, minimum SQ, and desired SQ related to the AS, IC, and LP. Along with these descriptive indicators, adequacy and superiority gaps were also calculated. The adequacy gap shows the difference between perceived SQ and minimum SQ, while the superiority gap indicates the difference between perceived SQ and desired SQ. The zone of tolerance is the difference between minimum SQ and desired SQ (Ahmed and Hossain Shoeb, 2009).

Table 3 shows that though the LSQ is above the mid-point (4.5) in every indicator, the perceived SQ is lower than both the minimum and desired levels of LSQ in most statements.

In the AS construct, overall perception scores are lower than the minimum and desired scores in the indicators except for the following three. The first was that the “library staff has the knowledge to answer users’ questions.” It remained slightly higher in perceptions ($M = 5.99$; $SD = 2.47$) than the minimum required level ($M = 5.98$; $SD = 3.68$). The second statement was, “Library staff is always willing to help users.” It also remained higher in perceived AS ($M = 5.91$; $SD = 2.47$) than the minimum acceptable level ($M = 5.85$; $SD = 2.47$). The third was that “Library staff shows dependability in handling users’ service problems.” It also remained higher in perceptions ($M = 6.25$; $SD = 2.43$) than the minimum required level ($M = 5.64$; $SD = 2.43$). This indicator exceeded the minimum acceptable and desired levels ($M = 6.21$; $SD = 2.50$). This shows that the staff is dependable beyond the desired levels.

The IC construct shows that the minimum acceptable level is higher than the perceived in six indicators. Only two indicators showed current quality higher than minimum acceptable level, i.e. (1) “the library’s website enables me to locate information on my own” ($M = 5.78$; $SD = 2.49$) than the minimum acceptable level ($M = 5.60$; $SD = 2.43$), and (2) “the library makes the information easily accessible for me” where perceived SQ was ($M = 5.94$;

Table 2. Demographics of the respondents

	Category no.	Frequency	Percentage	Missing
Gender	Male	209	52.3	–
	Female	190	47.5	–
	Total	399	99.8	1 (0.3%)
Area of residence	Rural	189	47.3	–
	Urban	211	52.8	–
	Total	400	100	0
School type	Government	165	41.3	–
	Private	229	57.3	–
	Total	400	100	0
Parent’s monthly income	Less than 25,000	107	26.8	–
	26,000 to 50,000	104	26.0	–
	51,000 to 75,000	98	24.5	–
	76,000 to 100,000	41	10.3	–
	More than 100,000	50	12.5	–
	Total	400	100	0

Source(s): Authors’ own work

Table 3. Perceived, minimum and maximum required LSQ by the school library users

		PSQ		MSQ		DSQ				
Affect of Service		Mean	Std. dev.	Mean	Std. dev.	Mean	Std. dev.	Adequacy gap	Superiority gap	Zone of tolerance
1	AS-1	5.95	2.565	6.10	2.557	6.21	2.503	−0.15	−0.26	0.11
2	AS-2	5.57	2.393	5.88	4.996	6.31	3.483	−0.31	−0.74	0.43
3	AS-3	5.84	2.359	6.12	2.586	6.20	2.430	−0.28	−0.36	0.08
4	AS-4	5.66	2.565	6.12	4.116	6.14	2.463	−0.46	−0.48	0.02
5	AS-5	5.99	2.472	5.98	3.689	6.09	2.464	0.10	−0.10	0.11
6	AS-6	5.77	2.503	6.01	2.426	6.39	2.411	−0.24	−0.62	0.38
7	AS-7	5.79	2.479	5.90	2.507	5.93	2.470	−0.11	−0.14	0.03
8	AS-8	5.91	2.473	5.85	2.479	6.23	2.469	0.06	−0.32	0.38
9	AS-9	6.25	2.437	5.64	2.431	6.21	2.503	0.61	0.04	0.57
Information Control										
10	IC-1	4.84	2.587	5.92	2.316	5.86	2.300	−1.08	−1.02	−0.06
11	IC-2	5.78	2.497	5.60	2.432	6.16	2.596	0.18	−0.38	0.56
12	IC-3	5.64	2.578	6.24	5.321	6.05	2.569	−0.60	−0.41	−0.19
13	IC-4	5.54	2.584	5.94	3.935	6.11	2.433	−0.40	−0.57	0.17
14	IC-5	5.80	2.623	5.99	2.431	56.29	2.512	−0.19	−0.49	0.30
15	IC-6	5.71	2.590	5.88	2.504	6.10	2.537	−0.17	−0.39	0.22
16	IC-7	5.94	2.419	5.69	2.539	5.81	2.365	0.25	0.13	0.12
17	IC-8	5.78	2.402	5.79	2.480	6.08	2.418	−0.01	−0.30	0.29
Library as a Place										
18	LP-1	6.44	2.485	6.20	2.437	6.33	2.429	0.24	0.11	0.13
19	LP-2	6.06	2.652	6.02	2.451	6.30	2.501	0.04	−0.24	0.28
20	LP-3	6.17	2.542	5.89	2.471	6.17	2.619	0.28	0.00	0.28
21	LP-4	6.27	2.473	6.00	2.467	6.56	5.206	0.27	−0.29	0.56
22	LP-5	6.08	2.436	5.77	2.380	6.25	2.437	0.31	−0.17	0.48

Source(s): Authors' own work

SD = 2.41) and minimum acceptable level was (M = 5.69; SD = 2.53). Except for these, the perceived SQ in IC is below the minimum acceptable level and needs improvement.

The LP construct exceeded the minimum acceptable level, but improvements are still required to meet the students' desired SQ. Therefore, perceived library quality exceeds the minimum acceptable library quality. In the LP construct, two indicators showed up to the mark quality. First, "the library has space that inspires study and learning" exceeds the perceived value (M = 6.44; SD = 2.48) by more than the desired value (M = 6.33; SD = 2.42). Another statement, "the library has a comfortable and inviting location," showed an equal score in the perceived (M = 6.17; SD = 2.54) and desired LSQ (M = 6.17; SD = 2.61). The visual images of the differences are displayed in radar maps. Building upon these findings, [Figure 1](#) visually

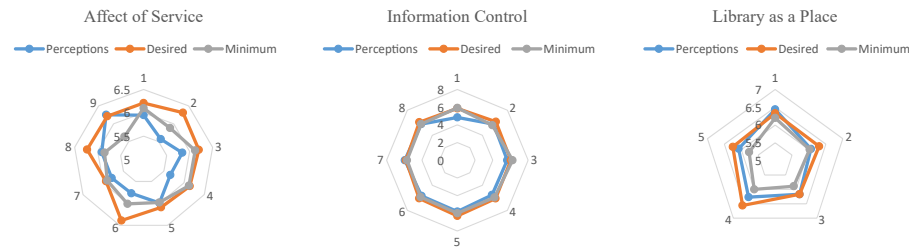


Figure 1. Radar map of difference in perceived, minimum, and desired SQ in AS, IC, and LP. Source: Authors' own work

presents the differences in perceived, minimum, and desired SQ in the realm of AS, IC, and LP, making results easier to understand.

Table 4 shows the details of *t*-tests applied to the perceptions, minimum acceptable, and desired library quality based on gender, school type, and background differences. The following text shows the finding in details. Levene's Tests for Equality of Variances revealed insignificant values in all three tests in the perceptions, all three tests in the minimum acceptable, and two test (gender and background based) in the desired library quality. However, the test revealed a significant value in the desired library quality based on the school type ($p < 0.04$). Therefore, the results of the *t*-tests were read in the first row (equal variances assumed) in all the former instances, and in the second row (equal variances not assumed) in the later instance of difference in the desired library quality based on school type.

The results of the *t*-test regarding perceptions of library SQ based on gender differences show that there were significant differences in perceived SQ ($p = 0.00$; $d = 0.29$; $\delta = 0.30$; $g = 0.29$) and desired SQ ($p = 0.01$; $d = 0.04$; $\delta = 0.05$; $g = 0.04$). This means that the male students perceived and desired SQ significantly higher than the females but the effect sizes in the perceptions were small and in the desired SQ were less than small. The minimum acceptable SQ was insignificant with a small effect size ($p = 0.61$; $d = 0.24$; $\delta = 0.24$; $g = 0.24$).

Similarly, the results of the *t*-test applied to gauge the differences in the perceived, minimum acceptable, and desired SQ based on school type showed that there was a significant difference in the minimum acceptable service quality ($p = 0.00$; $d = 0.07$; $\delta = 0.07$; $g = 0.07$). The government school students minimum acceptable SQ was significantly higher than that of private school students, though the effect size was small. However, there were no differences in perceived and desired SQ based on school type.

When interpreting the results of the third pool of *t*-tests applied to gauge the difference in all three constructs based on the background of the respondents showed no significant difference in the perceived ($p = 0.08$; $d = 0.16$; $\delta = 0.17$; $g = 0.16$); minimum acceptable ($p = 0.28$; $d = 0.11$; $\delta = 0.11$; $g = 0.11$); or desired SQ ($p = 0.28$; $d = 0.10$; $\delta = 0.10$; $g = 0.10$) based on background differences. The effect size also remained small, indicating small practical differences as small medium and large.

Table 5 shows the results of the One-Way-ANOVA test used to determine the variations in respondents' views related to currently perceived, minimum acceptable, and desired library quality depending on the five classified income groups. According to the data, most respondents were from the lowest income group, less than 25,000 rupees per month. Respondents whose parents earn between 76,000 and 100,000 rupees per month perceived the library services as the highest. The group whose parents earned more than 100,000 had the lowest minimum acceptable score. The highest library quality was required by the respondents whose parents earned 51,000 to 75,000 rupees per month. The results of a one-way ANOVA on the perceived level of school library services and minimum acceptable library quality based on the respondent's parents' income demonstrated a significant difference between different income groups ($p = 0.04$; $p = 0.01$) consecutively. Since it is a limitation of one-way ANOVA that it does not exactly indicate where the difference was found, a post hoc Tukey test was used to determine where the differences lay. The post-hoc-Tukey test yielded the following results:

Table 6 shows that there was a significant difference in perceived library quality between respondents whose parents earned between 26,000 and 50,000 ($M = 5.98$; $SD = 0.97$) and those whose parents earned more than 100,000 ($M = 5.49$; $SD = 1.11$; $MD = 0.49$). The respondents of former income brackets had perceived the library services significantly.

The table also shows that there was a significant difference in the minimum acceptable library quality between the respondents whose parents' income was less than 25,000 ($M = 6.26$; $SD = 0.1.30$) and those who had more than 100,000 ($M = 5.64$; $SD = 1.21$) with a mean difference of 0.62. The minimum acceptable library SQ was rated high by the respondents who belonged to former income bracket. The magnitude of difference in these groups was also very near to medium ($d = 0.493$; $\delta = 0.476$; $g = 0.487$).

Table 4. Differences in perceptions, minimum acceptable, and desired SQ based on gender, school type, and background

		Perceptions					Minimum acceptable					Desired				
		M and SD	Sig (2-tailed)	Cohen's <i>d</i>	Glass's <i>delta</i>	Hedges' <i>g</i>	M and SD	Sig (2-tailed)	Cohen's <i>d</i>	Glass's <i>delta</i>	Hedges' <i>g</i>	M and SD	Sig (2-tailed)	Cohen's <i>d</i>	Glass's <i>delta</i>	Hedges' <i>g</i>
Gender	Female = 190	M 5.74 SD 0.96	0.00	0.291	0.302	0.290	6.02 1.19	0.61	0.247	0.243	0.248	5.90 1.16	0.01	0.049	0.051	0.049
	Male = 209	M 6.03 SD 1.03					6.31 1.15					5.96 1.27				
School type	Pvt. = 229	M 5.84 SD 1.02	0.21	0.129	0.127	0.129	6.14 1.23	0.00	0.077	0.073	0.076	5.77 1.14	0.46	0.330	0.350	0.333
	Govt. = 165	M 5.97 SD 0.99					6.23 1.09					6.17 1.28				
Background	Urban = 211	M 5.81 SD 0.98	0.08	0.169	0.173	0.169	6.11 1.19	0.28	0.111	0.113	0.110	5.87 1.21	0.28	0.106	0.107	0.107
	Rural = 189	M 5.98 SD 1.03					6.24 1.15					6.00 1.22				

Note(s): *Sig. = < 0.05

Source(s): Authors' own work

Table 5. Differences in perceptions, minimum acceptable and desired library quality based on respondents' parents' income

	N	Perceived			Minimum acceptable			Desired		
		Mean	Std. dev.	Sig.	Mean	Std. dev.	Sig.	Mean	Std. dev.	Sig.
Less than 25,000	107	5.91	0.97	0.04	6.26	1.30	0.017	5.97	1.24	0.27
26,000 to 50,000	104	5.98	0.97		6.22	1.10		5.94	1.13	
51,000 to 75,000	98	5.94	1.02		6.28	1.03		6.09	1.05	
76,000 to 100,000	41	5.99	0.94		6.23	1.18		5.79	1.28	
More than 100,000	50	5.49	1.11		5.64	1.21		5.64	1.53	
Total	400	5.89	1.01		6.17	1.17		5.93	1.22	

Note(s): *Sig. = < 0.05

Source(s): Authors' own work

Significant differences were also found between those whose parents earned 26,000 to 50,000 ($M = 6.22$; $SD = 1.10$) and those who earned more than 100,000 ($M = 5.64$; $SD = 1.21$) per month with a mean difference of 0.58. The magnitude of difference in these group was medium ($d = 0.501$; $\delta = 0.527$; $g = 0.510$). Significant differences were also found between the respondents of the third income bracket whose parents earned 51,000 to 75,000 rupees per month ($M = 6.28$; $SD = 0.1.03$) and those who earned more than 100,000 rupees per month ($M = 5.64$; $SD = 1.21$) with a mean difference of 0.63. The magnitude of difference in these group was a bit higher than medium ($d = 0.569$; $\delta = 0.621$; $g = 0.585$). These two groups of lower income brackets also demonstrated that their minimum acceptable level of library quality was higher than the highest income bracket category. The results section showed the major results, inferences, and significant differences in the SQ of school libraries. The following discussion section discusses the finding in the context of existing literature.

Discussion

This study examines the SQ of School Libraries in seven Tehsils of Sargodha District. Four schools per Tehsil were taken as a sampling frame. The schools in each Tehsil comprised a government-sector boys' school, a government-sector girls' school, a boy's private sector school, and a girl's private sector school. SPSS 26 was used for data computations and analysis. Of the 400 survey respondents, 209 (52.3%) were males and 190 (47.5%) were females. The sample consisted of rural and urban students, with 229 (57.3%) studying in Private Schools and 165 (41.3%) in Government Schools.

Objective 1: status of perceived, minimum acceptable, and desired SQ in School Libraries

The perceived, minimum, and desired SQ scores were greater than the midpoint of 4.5 in each item. The perceived level of service was lower than intended when identifying the difference between perceived and desired library quality in most items (superiority gap). These results are consistent with the findings of [Kumar and Mahajan \(2019\)](#), who, using the LibQUAL+® tool, observed that the users' total perceived library SQ was lower than their desired level of LSQ. However, the perceived level of expertise, desire to assist users, and reliability of the library personnel in managing service difficulties are above the minimal necessary level. Earlier studies also discovered that the library personnel were adept at assisting patrons ([Mallya and Payini, 2016](#)).

The LP proved to be better than the AS and IC. The construct demonstrated that the values of all assertions in the perceived construct were above the minimum acceptable standards. These findings are consistent with a LibQUAL+® survey conducted at the University of Mississippi from 2001 to 2010 that determined that their sample's minimal library service standards were fulfilled and surpassed ([Greenwood et al., 2011](#)).

Table 6. Post-hoc-Tukey HSD comparisons of differences in perceived and desired library quality based on the respondents' parents' income

Multiple comparisons Tukey HSD												
(I) Parents' monthly income?		M	SD	(J) Parents' monthly income?	M	SD	Mean difference (I-J)	Std. error	Sig.	Cohen's <i>d</i>	Glass's <i>delta</i>	Hedges' <i>g</i>
Perceived SQ Minimum SQ	26,000 to 50,000	5.98	0.97	More than 100,000	5.49	1.11	0.49385*	0.17270	0.036	0.470	0.505	0.481
	Less than 25,000	6.26	1.30	More than 100,000	5.64	1.21	0.62164*	0.19982	0.017	0.493	0.476	0.487
	26,000 to 50,000	6.22	1.10	More than 100,000	5.64	1.21	0.58027*	0.20073	0.033	0.501	0.527	0.510
	51,000 to 75,000	6.28	1.03	More than 100,000	5.64	1.21	0.63949*	0.20272	0.015	0.569	0.621	0.585

Source(s): Authors' own work

Objective 2: difference in perceived, minimum acceptable, and desired SQ in the realm of Affect of Service; Information Control; and Library as a Place

The study's second objective was to know the status and differences in the perceived, minimum acceptable, and desired library quality in the realm of AS, IC, and LP. Although perceived SQ exceeded the midpoint value (4.5) across all cases, it remained lower than both minimum acceptable and desired SQ in most indicators. Three key measures were used to evaluate these gaps: those are adequacy gap, superiority gap, and zone of tolerance. The adequacy gap measures whether perceived SQ meets acceptable standard. A negative adequacy gap indicates that the services require improvement to meet the minimum acceptable standard.

On the other hand, the superiority gap indicates that whether the perceived SQ meets the desired quality or not. A negative superiority gap means that service require improvement to meet the desired library quality. Zone of Tolerance indicates a difference between minimum acceptable and desired SQ. A wide value indicates flexibility in expectations, and a smaller value suggest strict expectations.

Affect of Service. As mentioned earlier, all the indicators showed values higher than the mid-point (4.5) yet it remained below the both minimum acceptable and desired SQ in most of the indicators. The key improvements required were personalized attention (−0.46 adequacy gap, −0.48 superiority gap); staff expertise (−0.24 adequacy gap, −0.62 superiority gap); and library staff's consistent courtesy (−0.15 adequacy gap, −0.26 superiority gap).

The dependability of the library staff surpassed both measures (adequacy gap = 0.61; superiority gap = 0.04) indicating that the library users trust in the staff. However, there were few constructs that surpassed the minimum acceptable but required improvement in desired library quality. Those are library staff's willingness to help (adequacy gap = 0.06; superiority gap = −0.32); and staff's knowledge in answering the questions (adequacy gap = 0.10; superiority gap = −0.10). The zone of tolerance for most indicators was narrow meaning that library users have strict expectations regarding staff professionalism, attention, and expertise meaning they strictly require bridging the gap between their perceptions and desired expectations.

Library staff's dependability aligns with the findings of [Rehman \(2013\)](#), whose study also showed a positive adequacy gap (0.55). Similarly, [Kumar and Mahajan \(2019\)](#) found negative adequacy gaps in personalized attention, staff expertise, and courtesy. The narrow zone of tolerance is consistent with the earlier research study of [Rehman \(2012\)](#) in university settings (both ranging 0.25), indicating that users have strict expectations. Library staff's dependability aligns with the findings of [Rehman \(2013\)](#), whose study also showed a positive adequacy gap (0.55). Similarly, [Kumar and Mahajan \(2019\)](#) found negative adequacy gaps in personalized attention, staff expertise, and courtesy. The narrow zone of tolerance is quite consistent with the earlier research study of [Rehman \(2012\)](#) in the university settings (both ranging 0.25) indicating that users have strict expectations.

Information Control. In the realm of IC, most of the indicators revealed a negative adequacy and superiority gap, indicating that the perceived SQ needs to be improved both in minimum and acceptable levels. In this construct, although all the indicators scored more than the midpoint (4.5), yet majorly improvable indicators are the provision of electronic resources accessible from home (adequacy gap = −1.08; superiority gap = −1.02); the library's printed materials (adequacy gap = −0.60; superiority gap = −0.41); and ease of finding information at user's own (adequacy gap = −0.40; superiority gap = −0.57).

The IC showed positive in both indicators in the library making information accessible (adequacy gap = 0.25; superiority gap = 0.12). However, in the realm of library website enabling users to locate information, this indicator surpassed the minimum but required improvement to meet the desired level of service (adequacy gap = 0.18; superiority gap = −0.38). The zone of tolerance showed wider tolerance in some of the indicators while showed narrow in some. The timely updates and comprehensiveness of resources had a narrow zone of

tolerance, meaning that the users strictly required improvement in these up to reach their desired level.

The access to the digital resources is aligned with the research of [Pourahmad et al. \(2016\)](#) who found a -1.12 -adequacy gap in digital access. The superiority gaps in printed materials (-0.41) and finding information (-0.57) are aligned with [Ramezani et al. \(2018\)](#) whose resource availability showed a superiority gap of -0.50 . The results of [Asemi et al. \(2010\)](#) are however in contrast and broad with the present study's narrow zone of tolerance.

Library as a Place. In the LP, all indicators did not only exceed the midpoint (4.5) but also surpassed the minimum acceptable standard but required improvement to meet the desired SQ in the aspect of creating an inviting ambiance (adequacy gap = 0.04; superiority gap = -0.24); and fostering an environment conducive to enhance study and learning (adequacy gap = 0.27; superiority gap = -0.29); and providing spaces for collaborative learning (adequacy gap = 0.31; superiority gap = -0.17). However, the indicator of inspiring study spaces indicated positive in both measures (adequacy gap = 0.24; superiority gap = 0.11). The zone of tolerance was wider in this construct as compared to the other one suggesting that the library users are flexible in their expectations regarding the LP, though improvements are still required to improve the library SQ up to the desired level.

The superiority gap of ambiance and learning environment shows that these indicators do not meet users' expectations of the users. However, these results align with the study of [McCaffrey \(2013\)](#) meaning that Irish university libraries also do not meet their users' expectations as per their desires. The present results, however, are similar to the study of [Voorbij \(2012\)](#), who identified a wider zone of tolerance.

Objective 3: difference in perceptions, minimum and maximum expectations in the realm of Affect of Service; Information Control; and Library as a Place based on the type of school (government, private), gender (male, female), and socio-cultural status (urban, rural)

The uniqueness of the present study is that it is aimed at investigating unexplored School Libraries using the LibQUAL+® tool's Urdu version for the first time. Different inferential tests were applied since male and female students study in separate schools, belong to both urban and rural areas, and two different educational systems are offered to them: government sector schools and private sector schools. The study found that the perceived SQ significantly differed based on gender differences, with male respondents rating current library services significantly better than female respondents. The results of the present investigation contrast with the previous study conducted on a sample of Pakistani university library users by [Rehman \(2012\)](#). The researcher had not identified any significant difference in perceived SQ based on gender differences.

The study analysed students' perceptions of schools from different sectors, such as Private and Government Schools. The results showed no significant difference between these two groups, but the mean difference indicated that government school respondents perceived their library services comparatively better than private school respondents. There was no significant difference between students from urban and rural backgrounds in the realm of perceived library SQ.

The study analysed the desired library quality of respondents based on gender differences in both the Government sector and Private Schools and urban or rural backgrounds. Results showed that male students studying in Government Schools and those in rural areas had a higher desired library quality. The One-Way-ANOVA test showed that the respondents of 26,000 to 50,000 income brackets perceived the current library quality better than those from higher income brackets whose parents earned more than 100,000. All the lower income groups' minimum acceptable library SQ was higher than the highest income group.

Moreover, the respondents of the three lower income groups' minimum acceptable library quality was higher than the latter highest income group. The respondents of the low-income group perceived library quality as high, but their minimum acceptable level was also high.

Finally, the following section outlines key conclusions and suggestions based on the empirical investigations.

Conclusions

Overall, all the aspects of the library quality of all these three sub-constructs exceed the mid value. Therefore, it may be concluded that the perceived library quality in Sargodha district's private and Government Schools is better than the midpoint. When it comes to the difference between perceived SQ with the minimum (adequacy gap) and desired library quality (superiority gap), then a lot of improvements are required, ranging from AS to IC. Although the LP scored more than the minimum acceptable SQ, further improvements were needed to align with students' desired expectations.

The adequacy gap was positive and exceeded the minimum acceptable quality in only three statements in the AS, one in IC, and one in the AS Except for these five indicators, all others require improvement. The inferential analysis showed that a statistically significant difference based on gender differences was found in the perceived and desired library quality. The male respondents perceived and desired library quality significantly higher than the female students. The minimum acceptable SQ in Government Schools was significantly higher than in that of Private Schools. Moreover, the respondents who belong to the income bracket of 26,000 to 50,000 rupees perceived library quality significantly higher than those whose income exceeded 100,000 rupees. The minimum acceptable level of the lower income brackets was higher than those whose income exceeded 100,000 rupees.

Significance/rationale of the study

This research has theoretical and practical implications. Theoretically, it addresses a gap in global and national literature by validating and ensuring the reliability and validity of the Urdu version of the LibQUAL+® instrument by reducing interpretation and translation biases on a sample of school students. Moreover, it provides an empirical evidence on school library SQ by giving insights into gender-based and school types differences. It empirically identifies the adequacy gap, superiority gap, and zone of tolerance between perceptions, minimum acceptable, and desired library quality. These findings are crucial for school librarians, administrators, and educators designing continuous education programs. These findings may help them target gaps and improve library quality to desired levels. Librarians can use these insights to enhance services independently, while administrators and principals can support these improvements.

Additionally, continuous education program designers can develop training programs to bridge identified gaps. Overall, the empirical suggestions provide actionable insights for all stakeholders for the improvement of school library services. The practical areas of actions for the school administrators and policymakers are as follows:

Recommendations. The following recommendations are being furnished to the school librarians, administrators, and policymakers to make policies and take action:

Related to the Affect of Services.

- (1) School libraries staff must build confidence in the School Libraries users
- (2) School libraries staff must give every library user individual attention
- (3) The School Libraries staff must show courtesy and politeness
- (4) The School Libraries staff must always remain ready to respond to the queries and questions of the library users
- (5) The School Libraries staff must have knowledge to answer the library users' questions
- (6) The School Libraries staff must treat their library users with care and respect

- (7) The School Libraries staff must understand their users' needs
- (8) The School Libraries staff must always remain ready to help their users.

Related to Information Control.

- (1) The School Libraries electronic resources must be accessible from home
- (2) The School Libraries website must allow students to locate information on their own
- (3) The School Libraries must have printed materials that students require for their work
- (4) The School Libraries must have electronic information resources that students require
- (5) The School Libraries must have modern equipment that lets students easily access the needed information
- (6) The School Libraries must have easy-to-use modern tools that allow library users to find their required information on their own
- (7) The School Libraries must have a collection of electronic and print-format journals.

Related to Library as a Place.

- (1) The School Libraries must be quiet spaces where their users may study with concentration
- (2) The School Libraries must be converted to places that are gateways for study, research, and learning
- (3) The School Libraries must be transformed into places for discussion, group studies, and group learning.

General suggestion based on inferential analysis.

- (1) Female students must be provided with the orientations to the library and its services
- (2) The Private Schools' librarians and administration must improve their libraries as their counterparts have shown significantly better SQ.

Limitations of the study

The study utilized an equal-sized convenience sampling technique across the four schools in each Tehsil of Sargodha District. Therefore, there are limitations to convenience sampling regarding generalizability. The findings of this investigation may not be fully generalizable to all School Libraries in Pakistan.

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